Report_DEB

1. Importation des données

```r
file_path <- here::here("mod/J_DEB_SAAWBe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```
NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 =
seeds <- c(C1 = 3333, C2 = 3663, C3 = 2212)

text_priors <- '
Distrib (Fm,           TruncNormal,           0.1,      0.2,  0.01,      0.8);
Distrib (pAm,          TruncNormal_cv,        100,      0.2,    40,    2000);
Distrib (r_pAm_pM,     TruncNormal_cv,        0.8,      0.1,    0.1,      2);
Distrib (v,            TruncNormal_cv,    0.18505,      0.1,  0.001,    2);
Distrib (kap,          Uniform,               0,     1);
Distrib (kapA,         Uniform,               0.1,     0.9);
Distrib (Ehp,          TruncNormal_cv,         100,      0.2,    40,    300);
Distrib (E_coc,        TruncNormal_cv,          30,      0.2,     1,    300);
Distrib (rOM_ClxHorse, LogUniform,         0.00001,      0.3);
Distrib (b,            Uniform,            1,        20);
Distrib (dv,           TruncNormal_cv,           2,      0.5,     0,     10);
Distrib (wE,           LogUniform,      0.000000001,     0.1);

# Calcul des vraissemblances
Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,       Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIte

# ./mcsim.DEBcali DEB_C1.in
```

`OM_horse` and `OM_soil` distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

```
                  Fm.1.    pAm.1. r_pAm_pM.1.       v.1.     kap.1.    kapA.1.
```

```
Result_mode 0.52641400 143.27700  0.83670900 0.18217900 0.94919300 0.73515600
2.5%        0.38977200 115.63100  0.75489400 0.15214900 0.93421300 0.62955900
97.5%       0.57518625 172.64000  1.02520850 0.21535755 0.96794795 0.81873300
Min         0.10625500  94.58760  0.65133000 0.13310000 0.50515300 0.18063900
Max         0.67893100 186.42100  1.12724000 0.24555900 0.97365600 0.89182600
Result_mean 0.47752318 143.65487  0.88594863 0.18406393 0.95005794 0.72159592
Result_sd   0.04790535  13.86525  0.07023458 0.01599875 0.00929446 0.04866586
               Ehp.1.  E_coc.1. rOM_ClxHorse.1.       b.1.     dv.1.
Result_mode 111.22500 24.973700    0.0035904600  1.6047800 1.8648100
2.5%         83.13040 17.441400    0.0021988360  1.0440145 1.4495100
97.5%       144.41300 37.016435    0.0063575865  2.6921415 2.5307000
Min          69.81510 12.138900    0.0000161998  1.0002400 0.5220400
Max         167.71600 46.776600    0.0103937000 10.3284000 3.2171700
Result_mean 111.38811 27.231663    0.0040543908  1.6910517 1.9475950
Result_sd    15.57086  4.832912    0.0010758339  0.4397663 0.2827646
                   wE.1. Sigma_W.1.
Result_mode 1.173550e-05 0.10665700
2.5%        8.519846e-06 0.09735700
97.5%       1.576533e-05 0.12405400
Min         1.240480e-09 0.08563710
Max         1.955330e-05 5.72094000
Result_mean 1.148369e-05 0.11033738
Result_sd   2.031010e-06 0.05375568

Number of observations, n =  277
Number of parameters, k =  12
Log-likelihood, LL =  3.293537
BIC =  60.90114


Potential scale reduction factors:

               Point est. Upper C.I.
b.1.                 1.49       2.42
dv.1.                1.12       1.37
E_coc.1.             1.07       1.21
Ehp.1.               1.06       1.09
Fm.1.                1.01       1.04
kap.1.               1.17       1.49
kapA.1.              1.11       1.31
pAm.1.               1.39       2.11
r_pAm_pM.1.          1.02       1.04
```

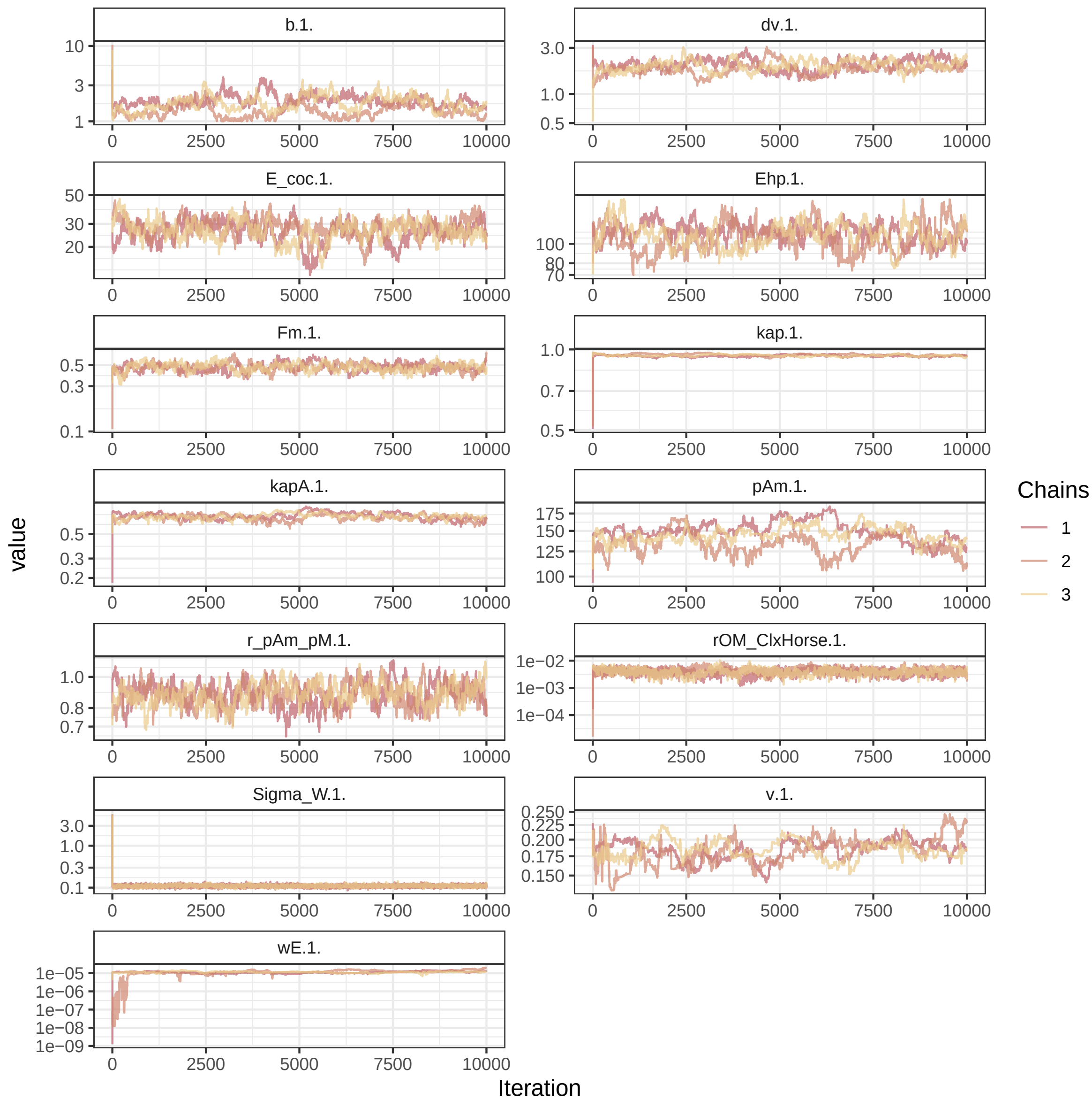


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

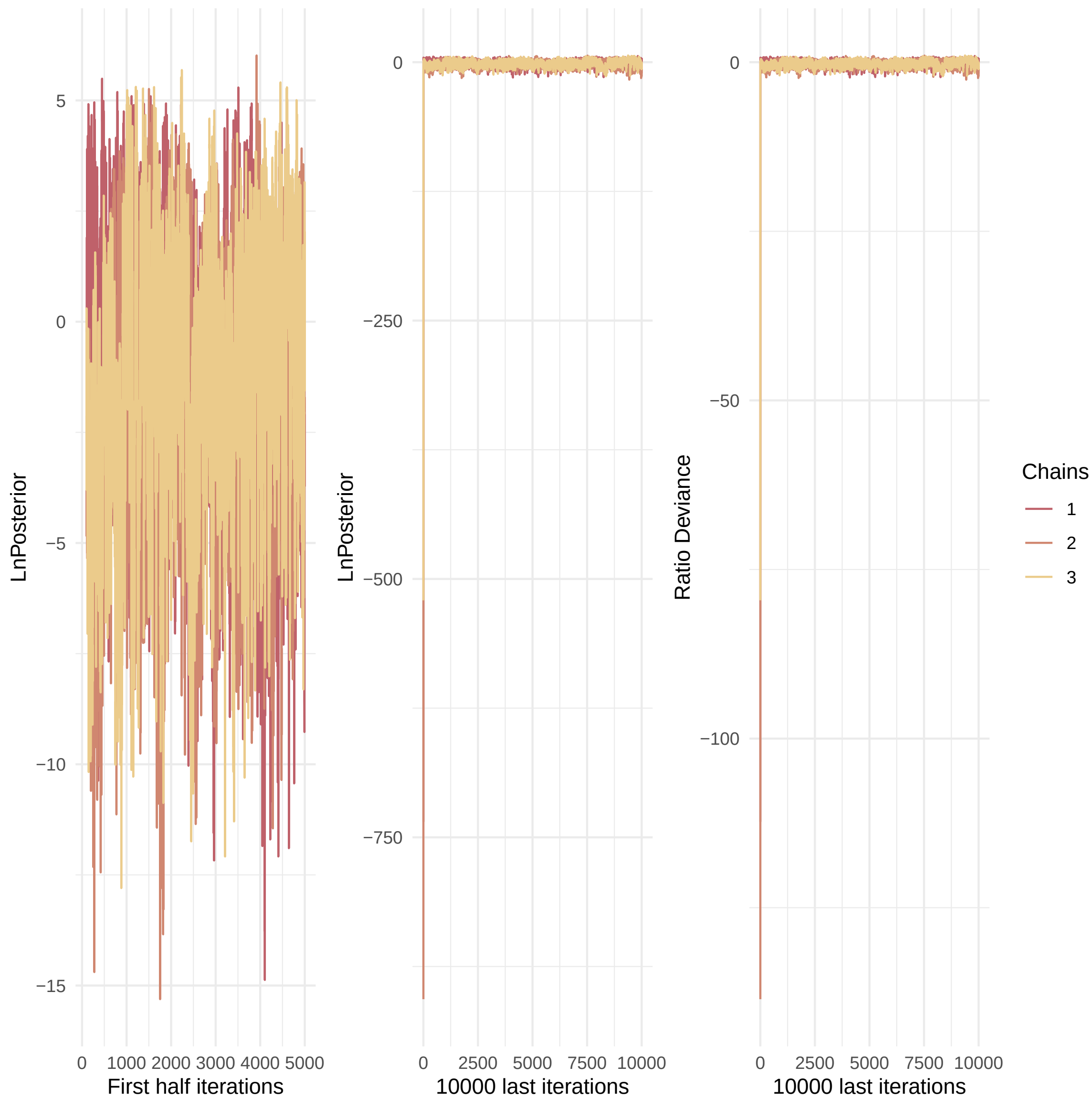


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

```
rOM_ClxHorse.1.       1.08       1.26
Sigma_W.1.            1.01       1.02
v.1.                  1.19       1.55
wE.1.                 1.51       2.47

Multivariate psrf

1.64
```

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE])  # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

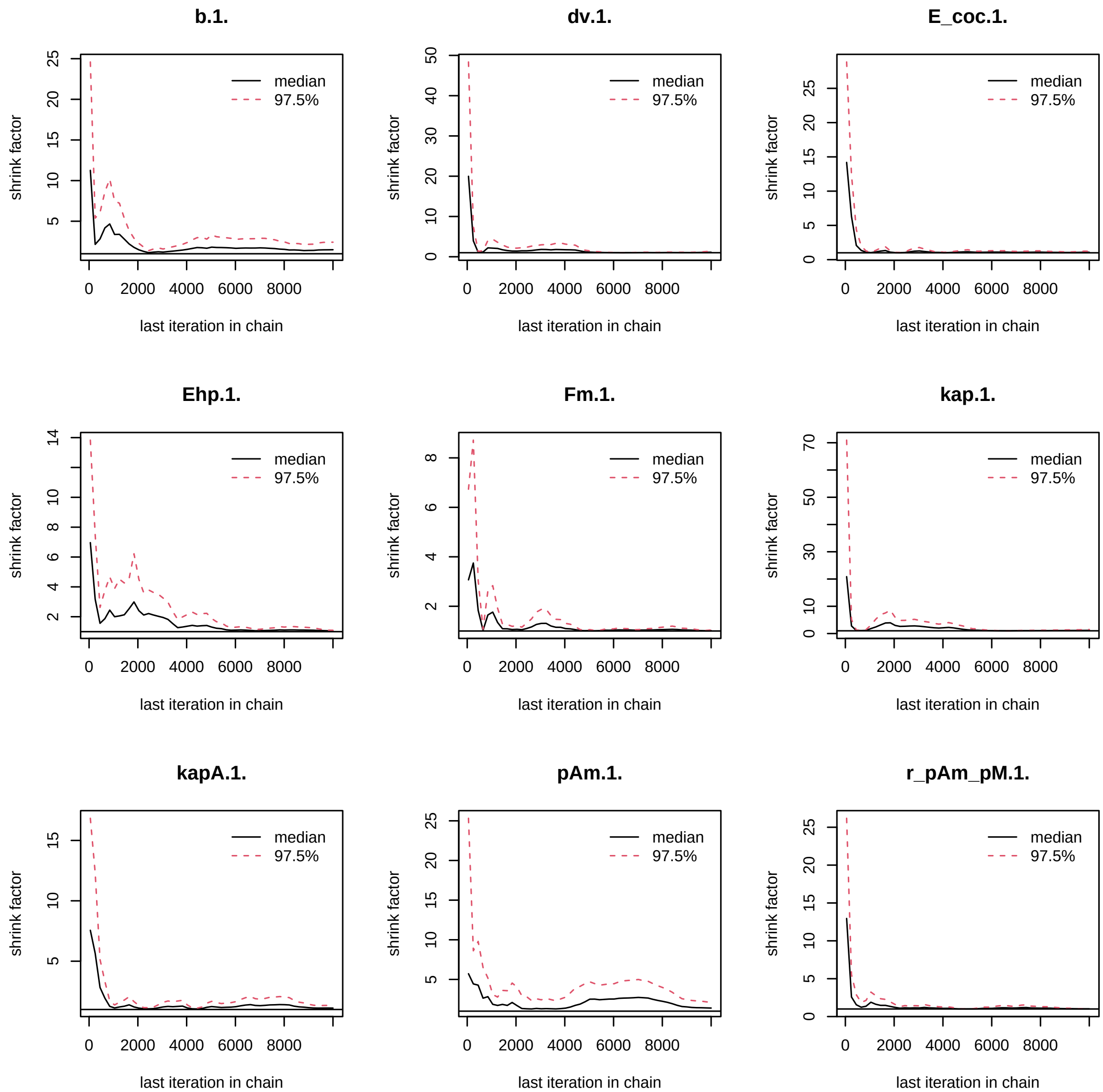


Figure 3: Chains convergence.

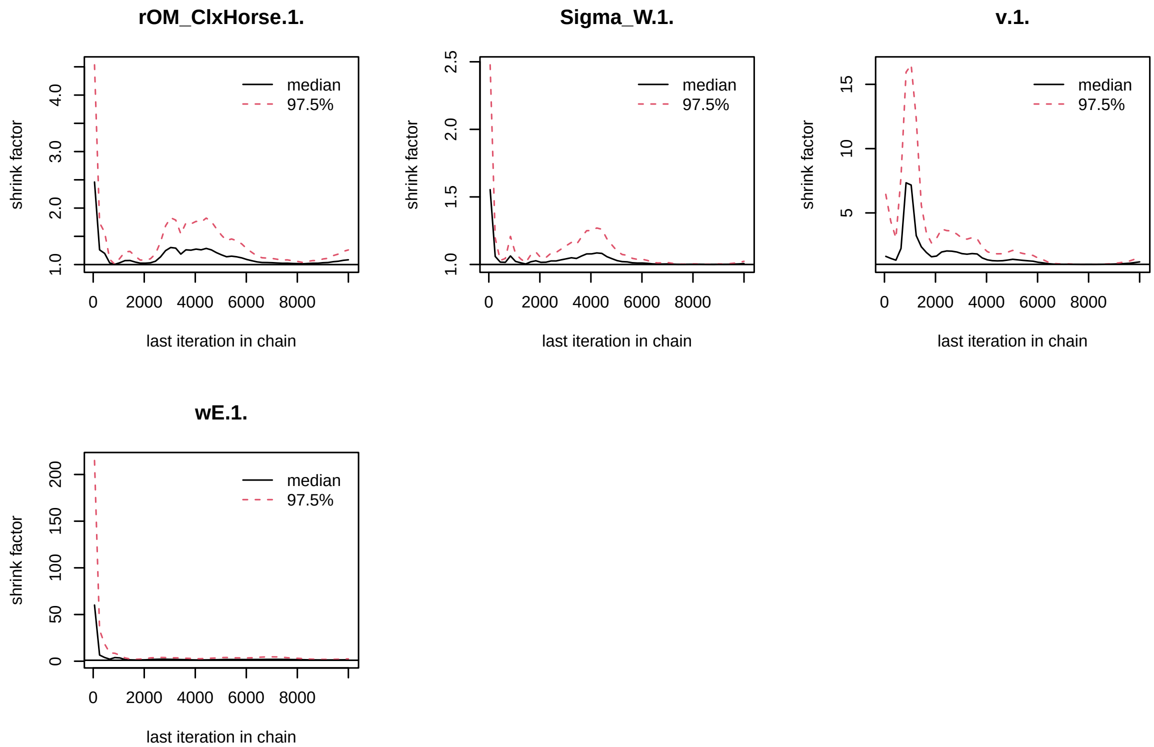


Figure 4: Chains convergence.

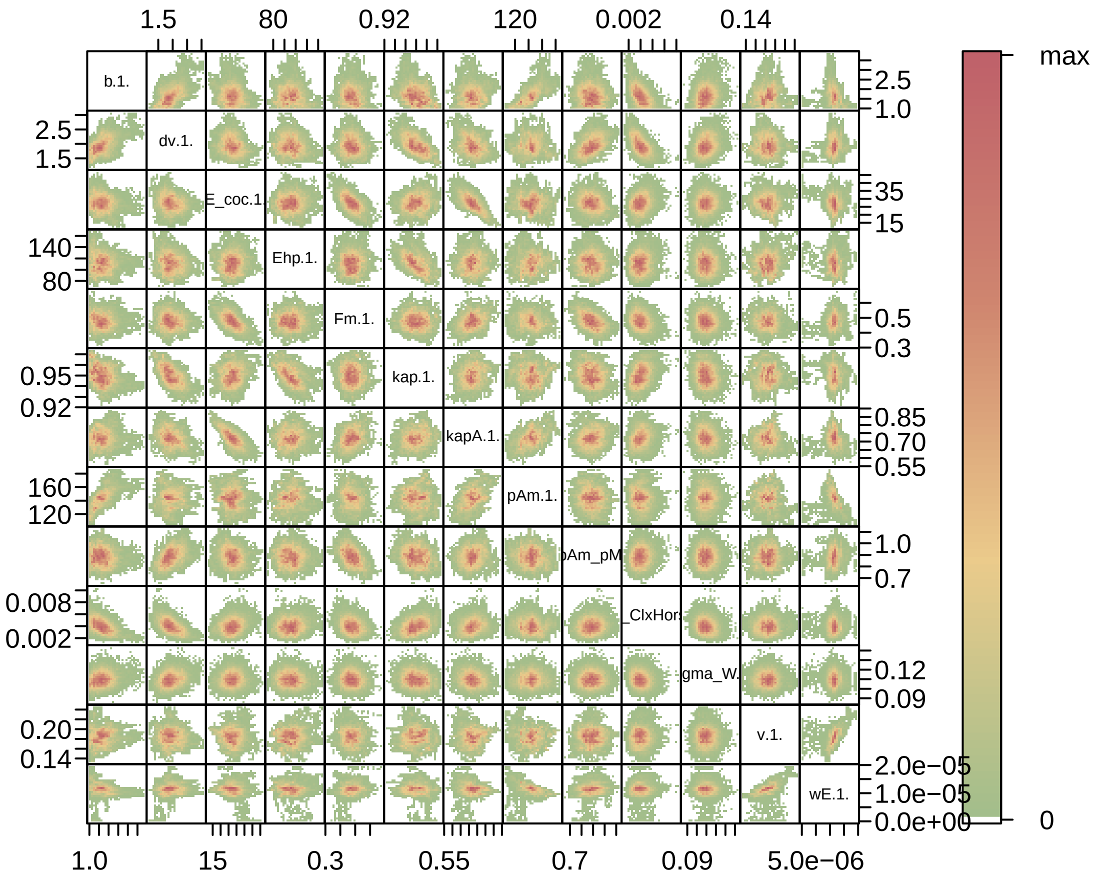


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

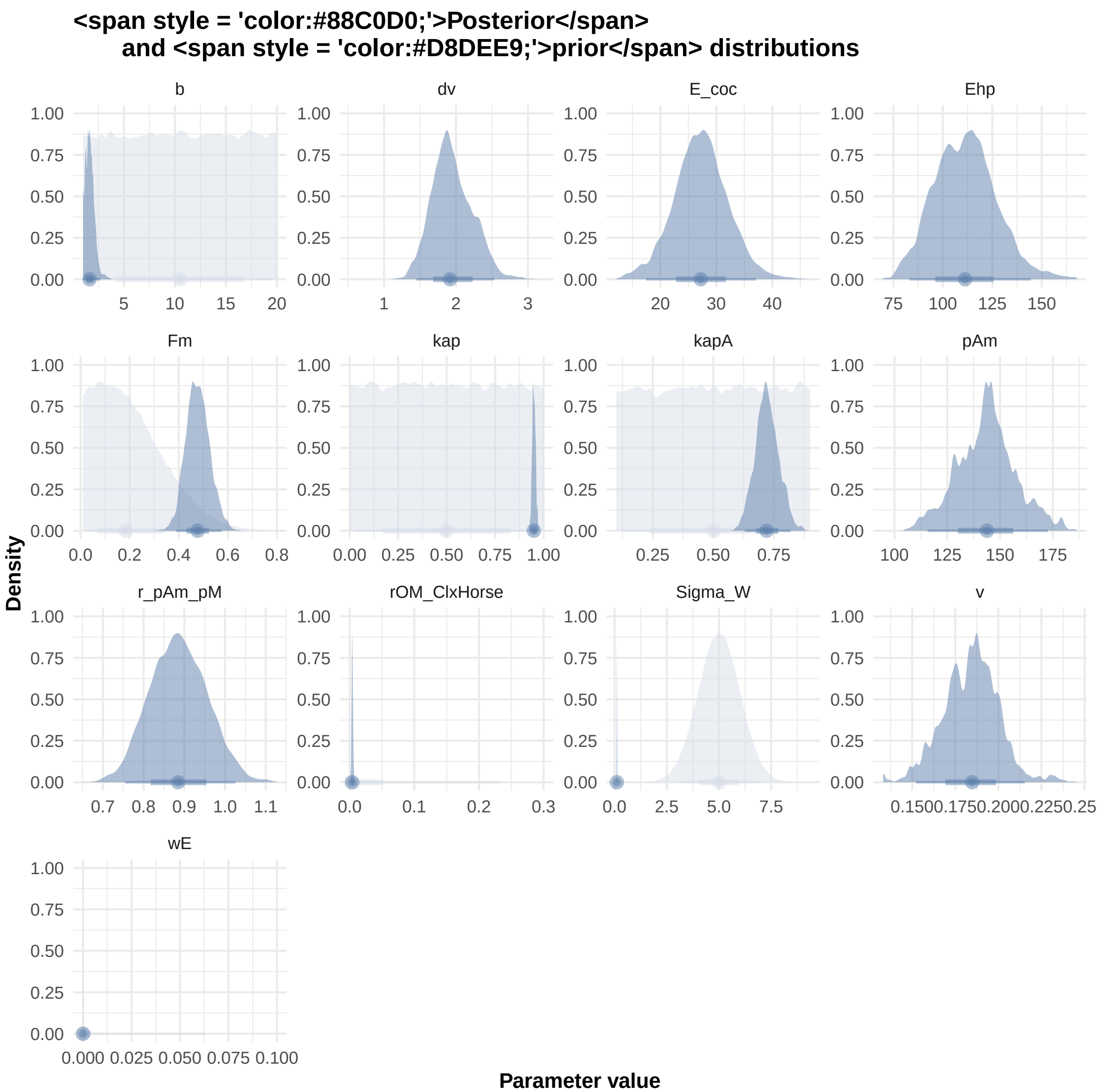


Figure 6: Distributions of the priors and the posteriors.

```
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.$|\\.1\\.", "", param),
    line  = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
    Length_struct,
    Reproduction,
    Energy,
    Organic_matter,"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_prin
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcali DEB_sim_full.in
```

```
##############################
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
```

```
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#############################

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
```

```
Warning: `read_table2()` was deprecated in readr 2.0.0.
i Please use `read_table()` instead.
```

```
Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpe
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

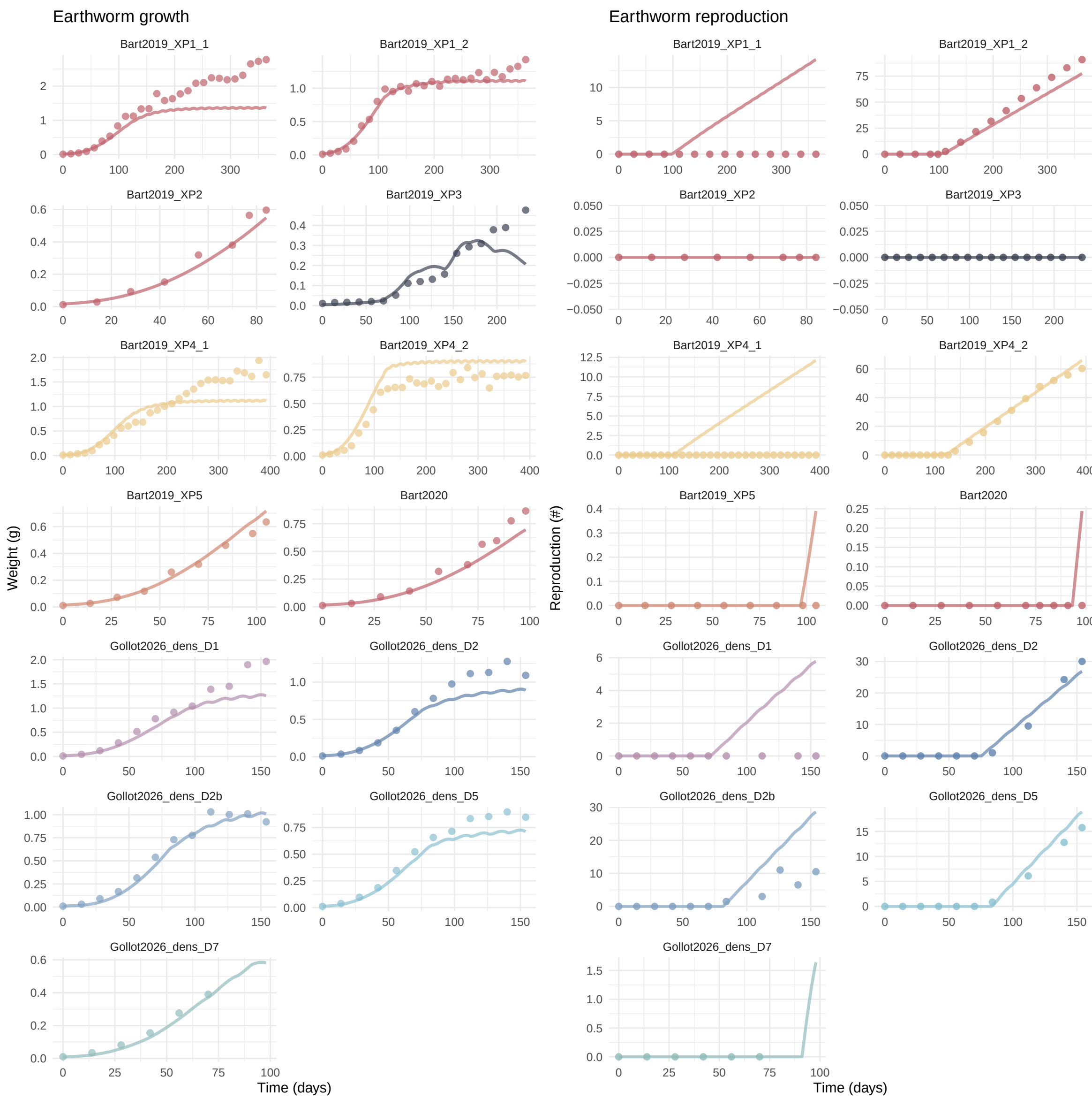
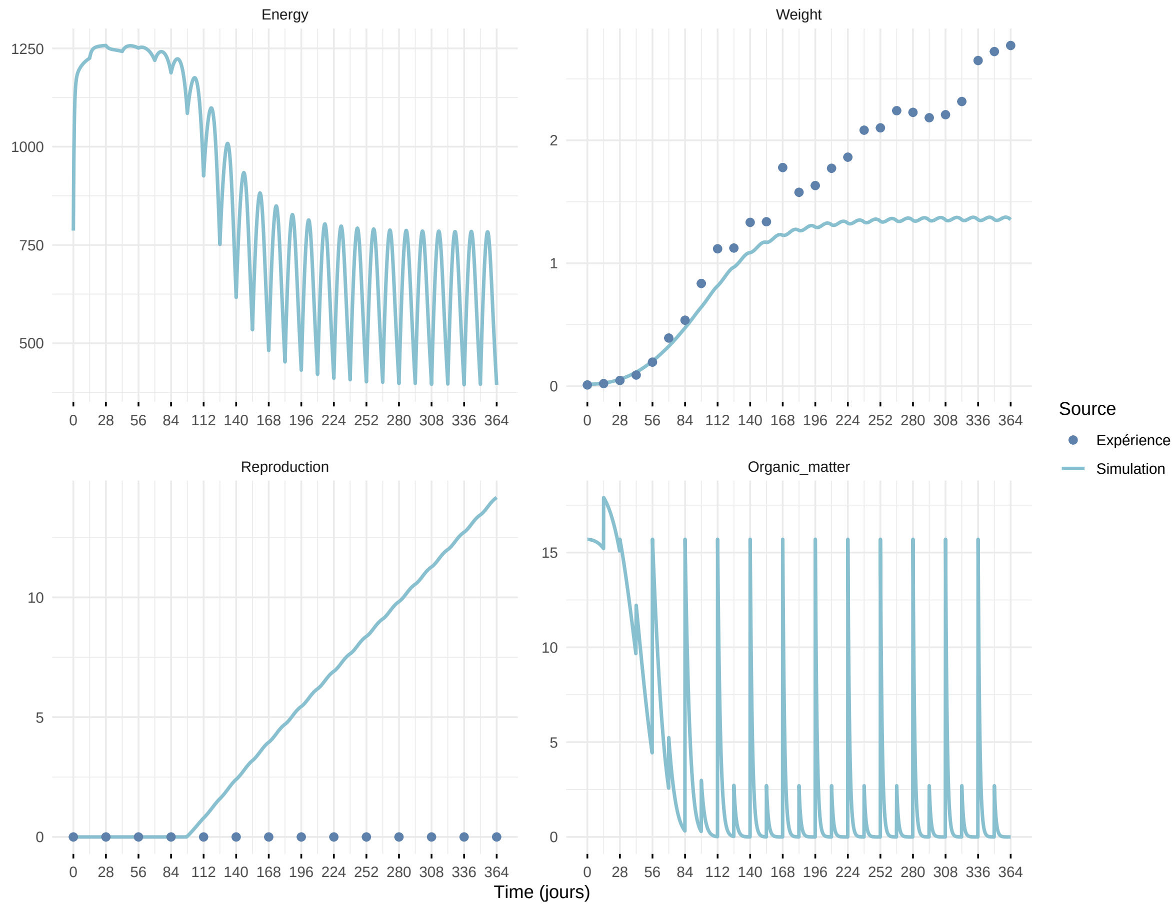
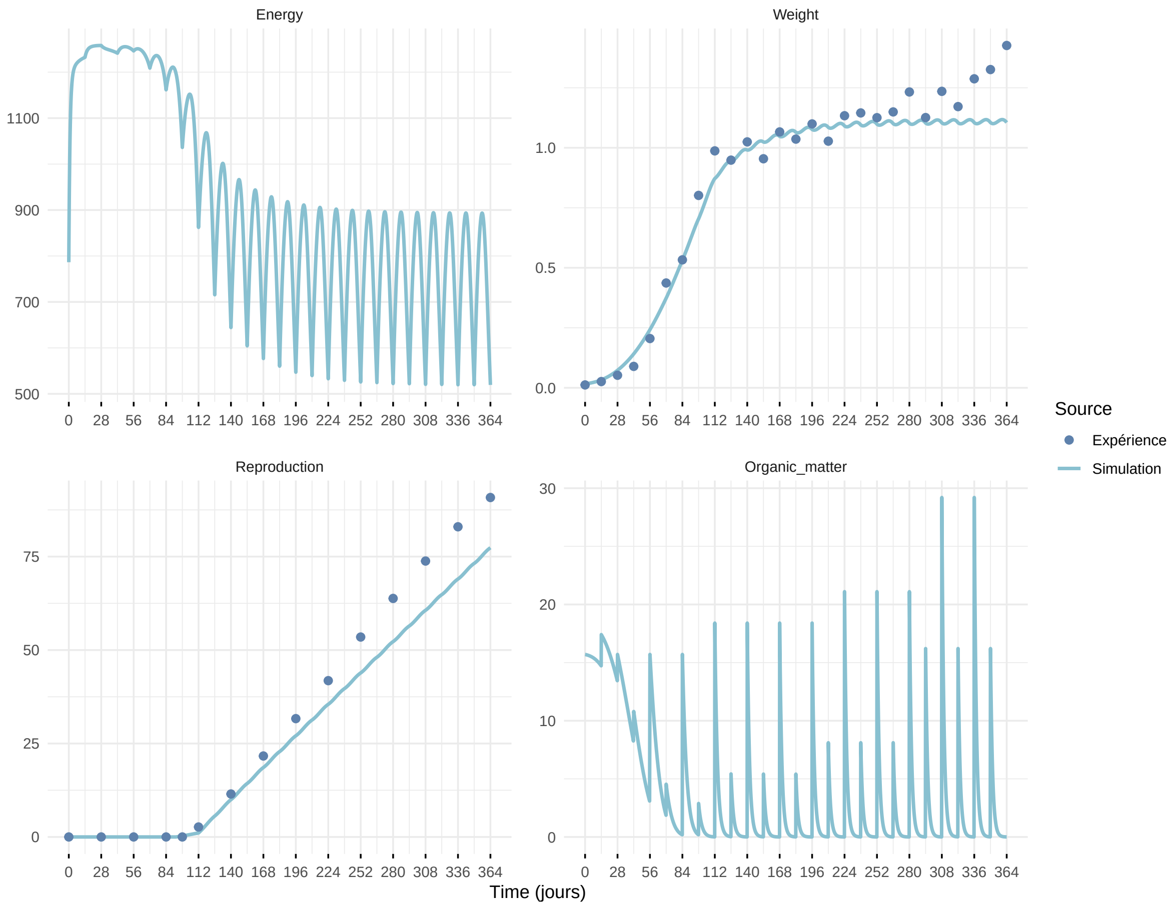


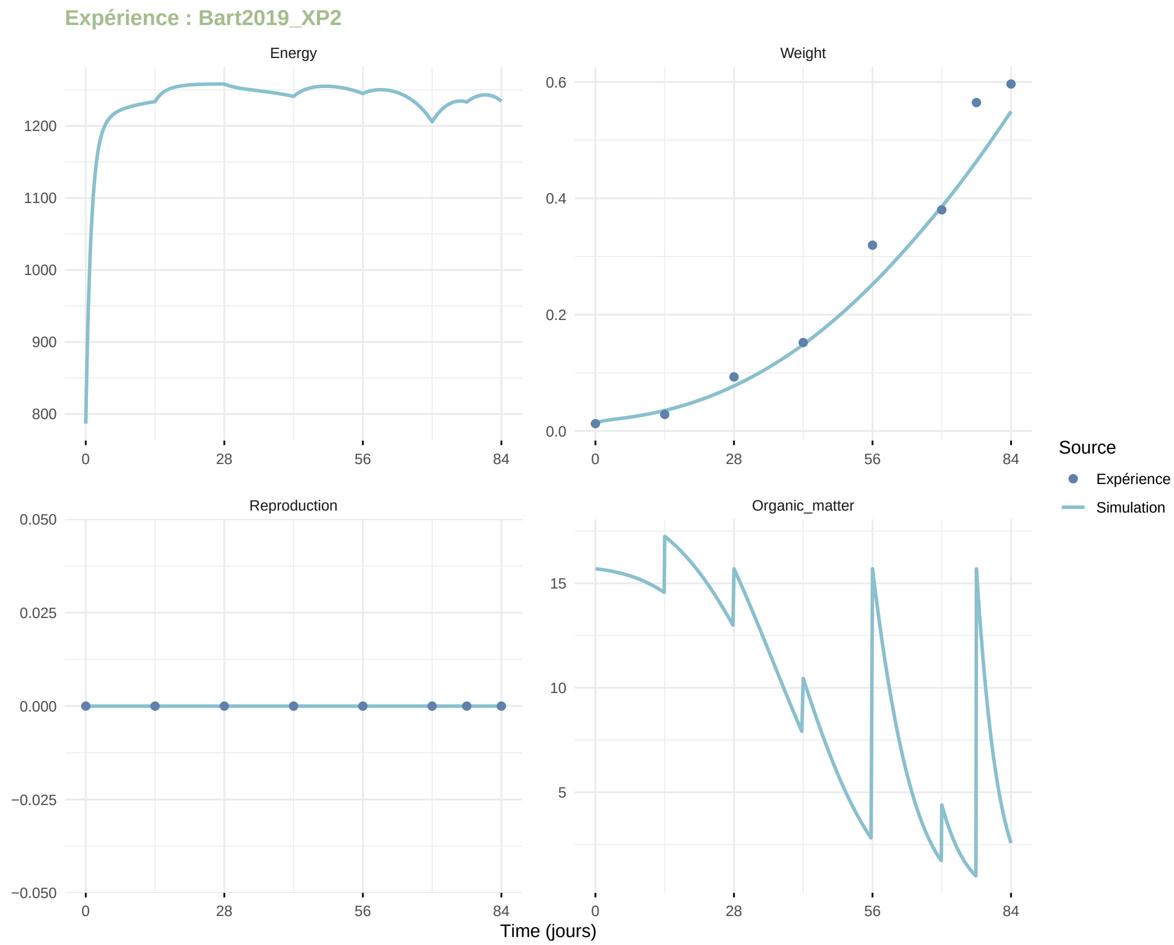
Figure 7: Simulations Weight and Reproduction.

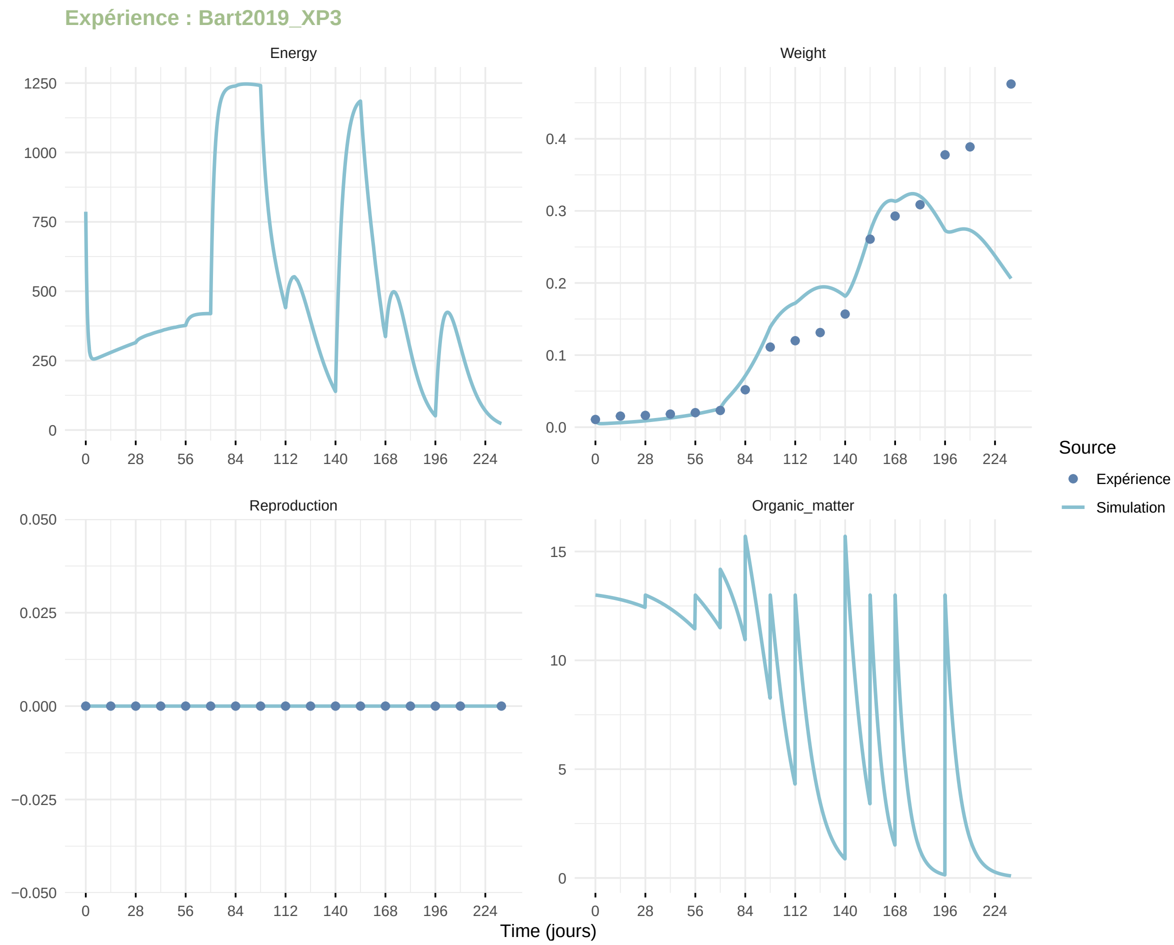
Expérience : Bart2019_XP1_1



Expérience : Bart2019_XP1_2

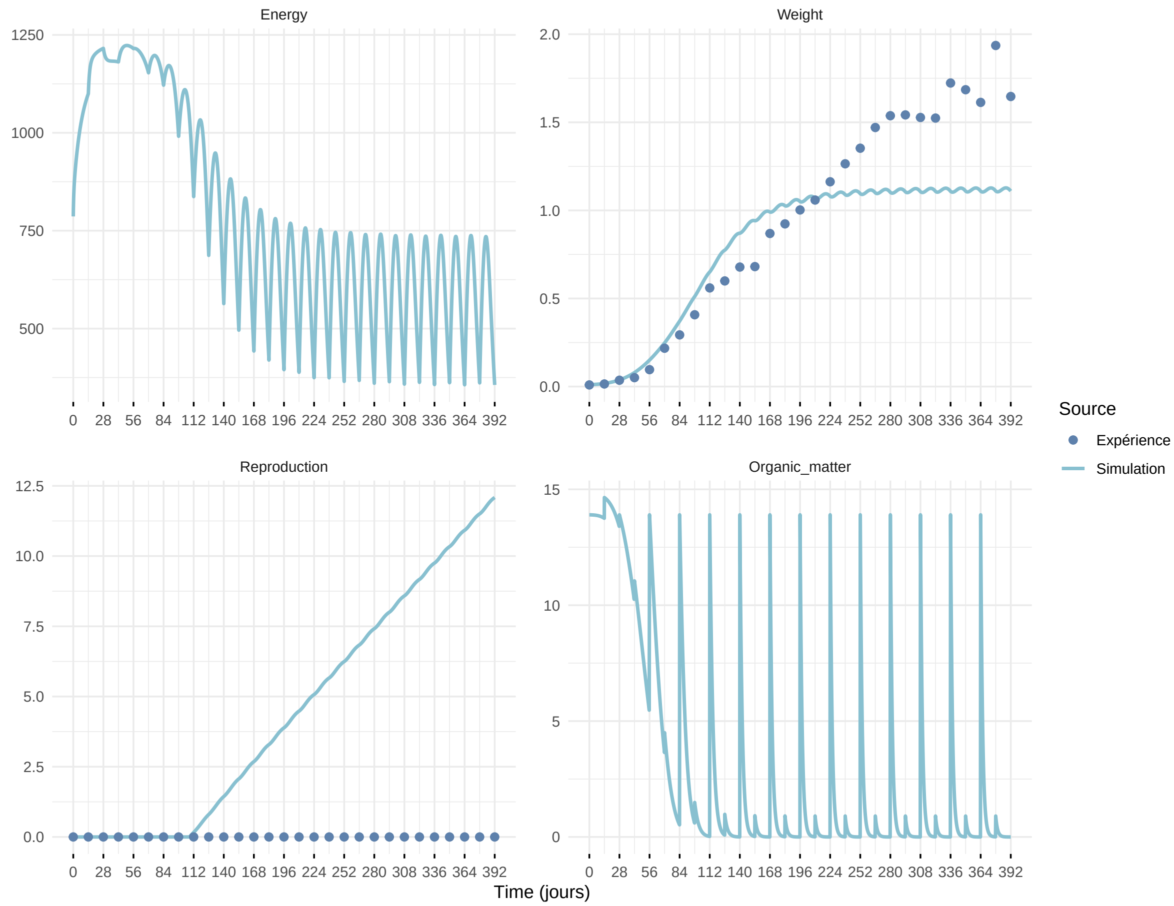



Expérience : Bart2019_XP2
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

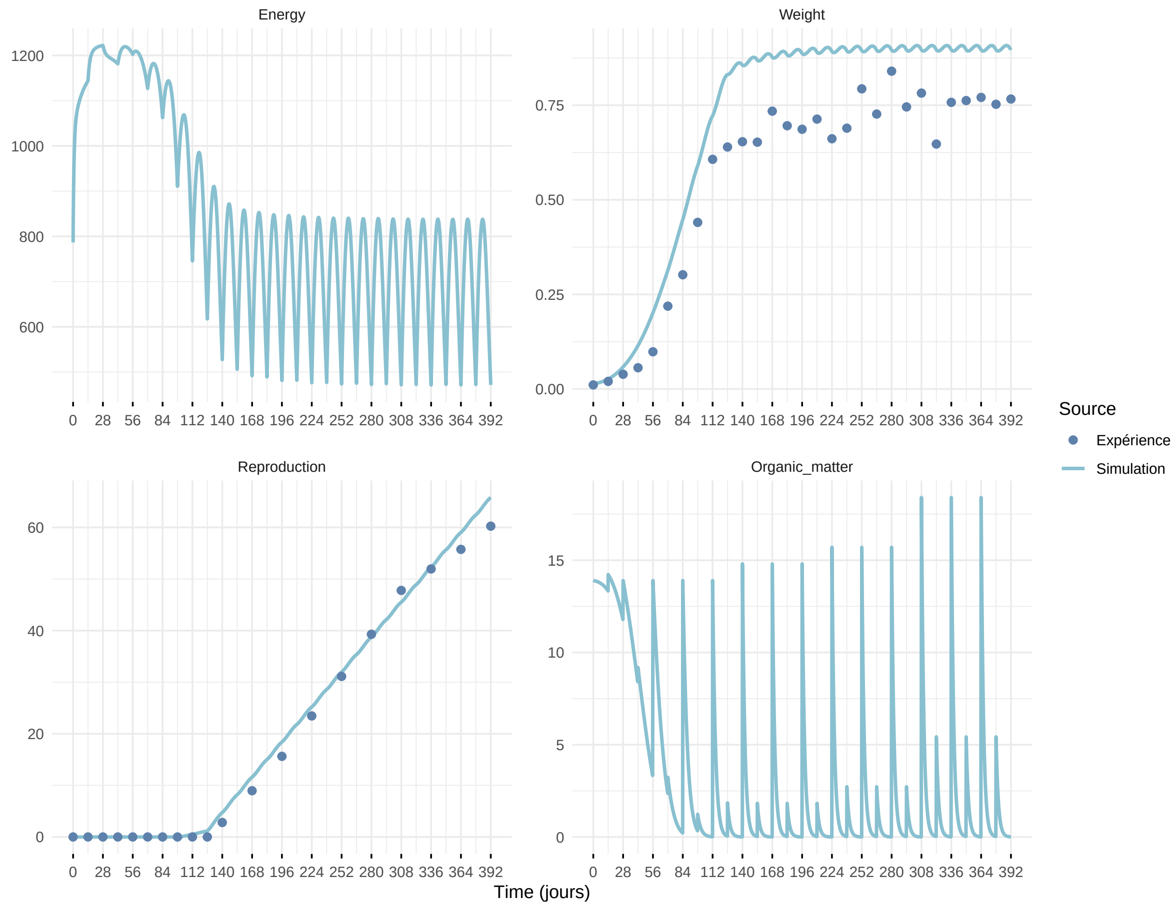


Expérience : Bart2019_XP3
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

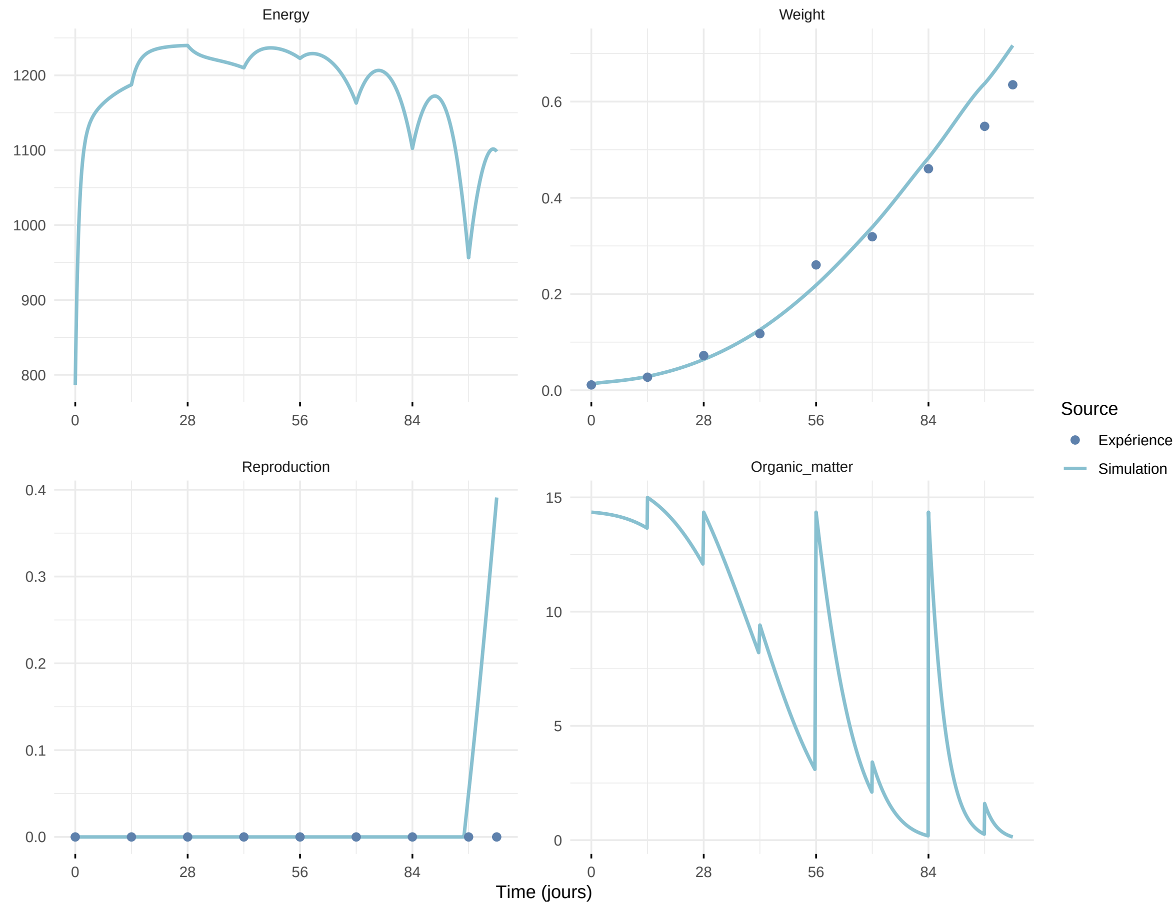
Expérience : Bart2019_XP4_1



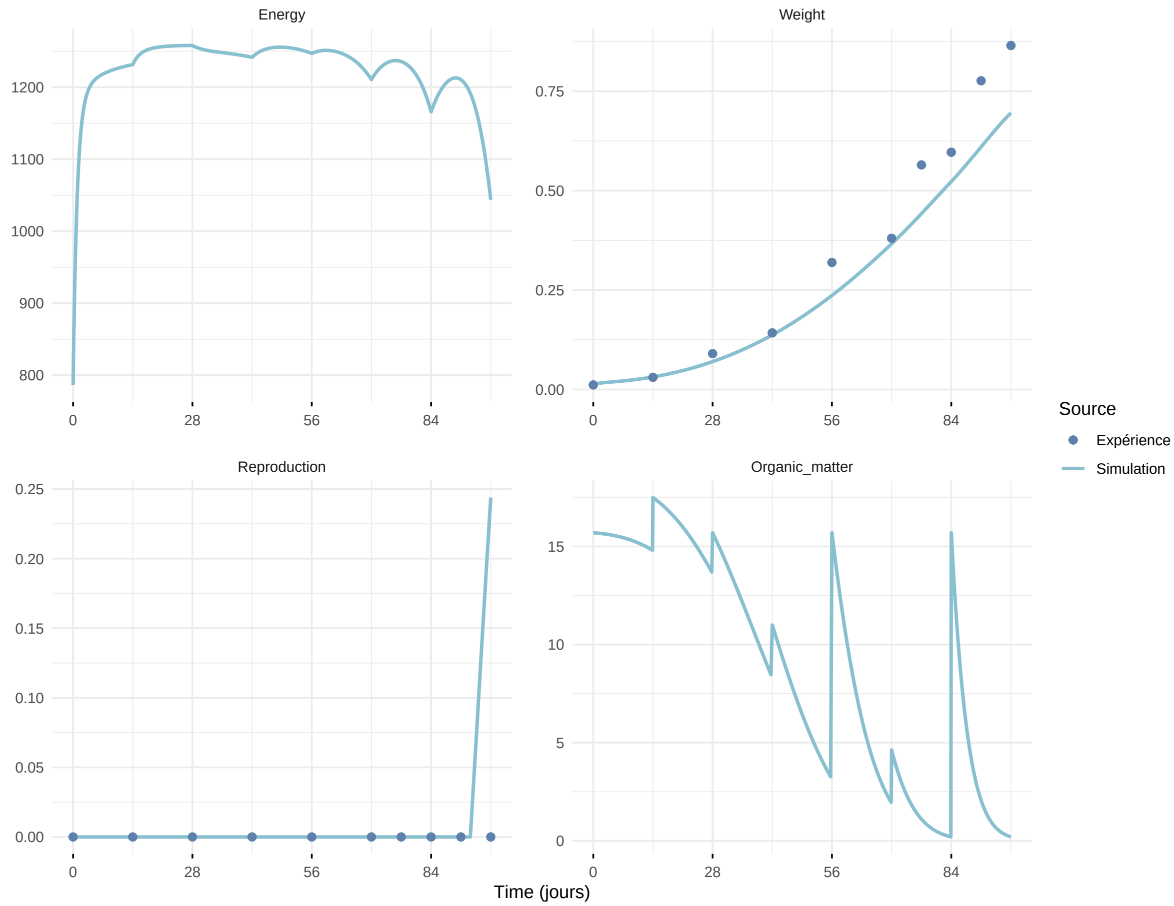
Expérience : Bart2019_XP4_2



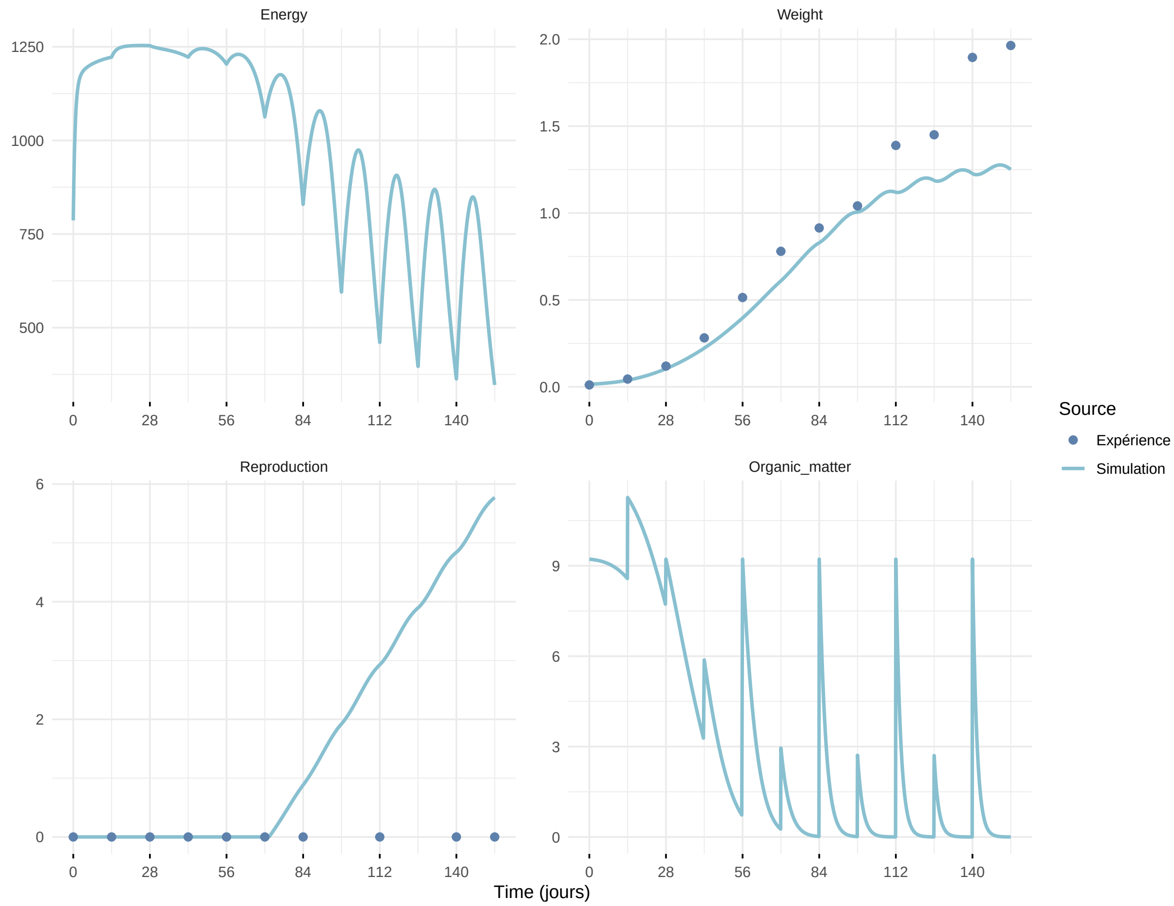
Expérience : Bart2019_XP5



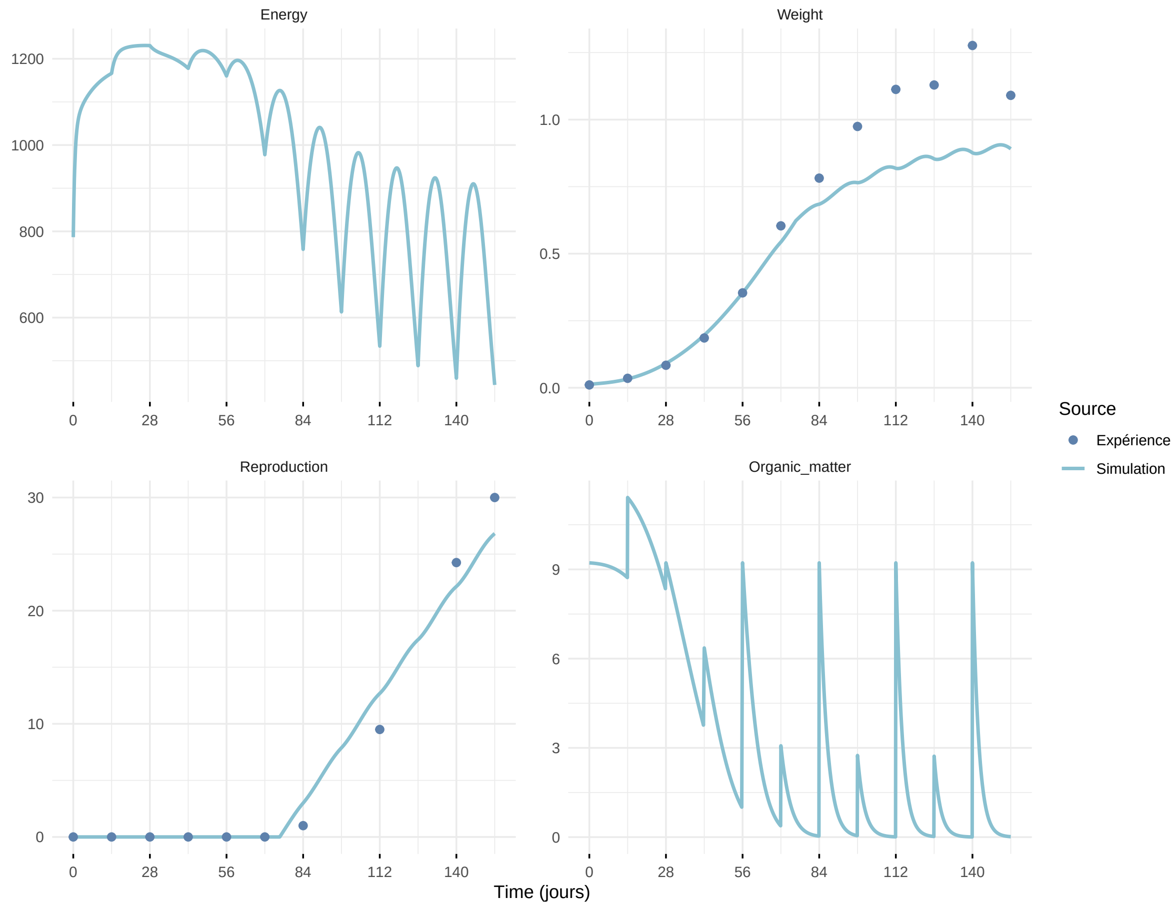
Expérience : Bart2020

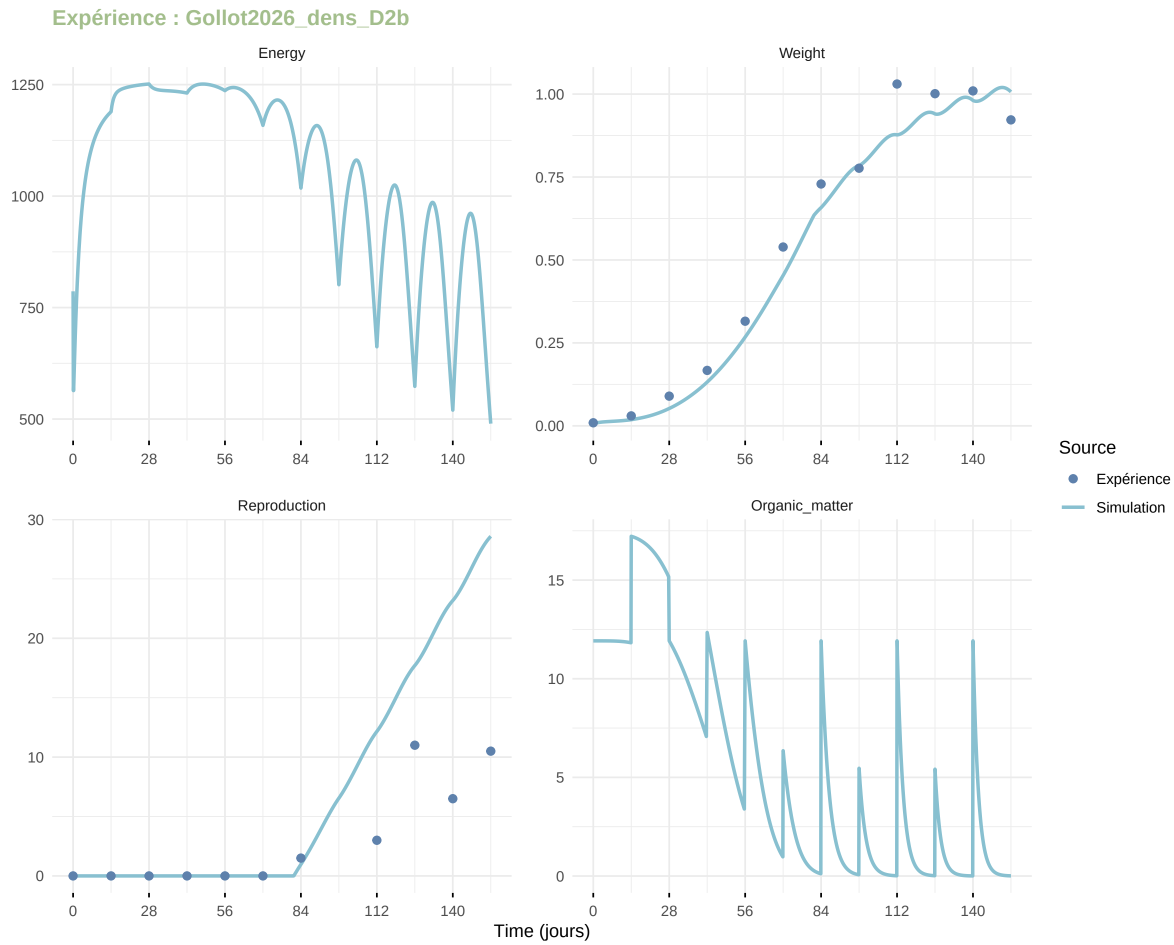


Expérience : Gollot2026_dens_D1

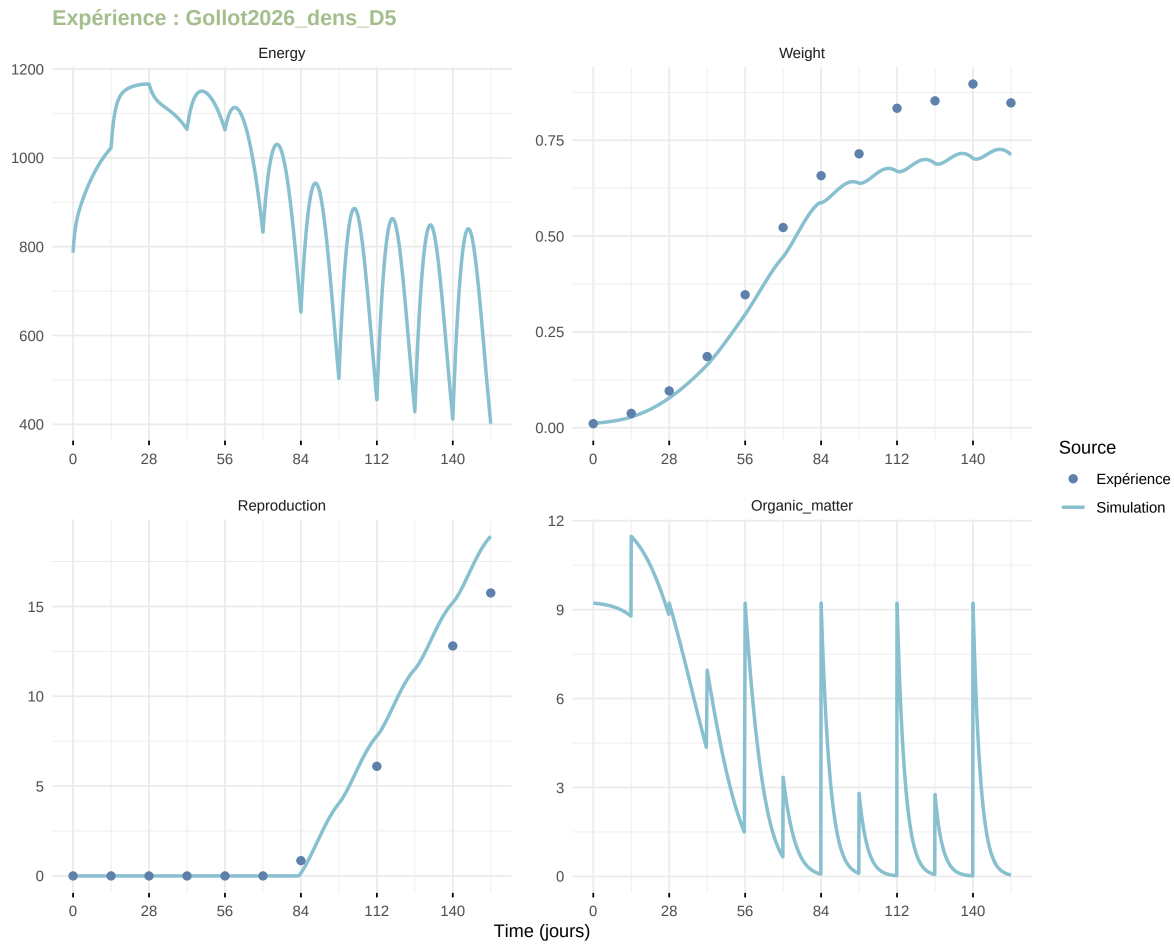


Expérience : Gollot2026_dens_D2





Expérience : Gollot2026_dens_D2b
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)


Expérience : Gollot2026_dens_D5
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)
1200
1000
800
600
400
0.75
0.50
0.25
0.00
15
10
5
0
12
9
6
3
0
28
56
84
112
140

Expérience : Gollot2026_dens_D7